

Highlights

Conformal Surprisal Fusion for Detecting Prompt-Injection Compromise in LLM Agent Traces

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- Fuses observation- and action-side trace signals by conformal surprisal
- Raises AUROC 0.937 to 0.953; the gain in stability is the more robust effect
- Outranks a re-implemented Task Shield run on the same traces, 0.953 to 0.764
- Subtracting the benign maximum provably caps AUROC at one half
- AgentDojo hides almost no compromise, so it cannot test the hard case